

Best Practices in Corpus Linguistics

What lessons should we take from the Replication Crisis and how can we guarantee high quality in our research?

Dr. Martin Schweinberger

slides available at www.martinschweinberger.de



THE UNIVERSITY
OF QUEENSLAND
AUSTRALIA

CREATE CHANGE

Aims, definition, and the current state of affairs

What are best practices?

A **best practice** is a method or technique that is superior to alternatives because it produces results that are more reliable, transparent, replicable, and in compliance with legal or ethical requirements.

Aims, definition, and the current state of affairs

This presentation aims to

- Raise awareness for *Best Practices* in Corpus Linguistics
- Discuss issues related to Best Practices and Replicability
- Propose improvements to current research practices
- Offer solutions on how best practices can be implemented

Aims, definition, and the current state of affairs

Best practices have come into focus as a result of the **Replication Crisis** (RC) which is a ongoing methodological crisis primarily affecting parts of the social and life sciences beginning in the early 2010s.

Nature 2016 poll of 1,500 scientists:

- 70% failed to reproduce at least one other scientist's experiment
- 50% failed to reproduce one of their own experiments (Fanelli 2009)

Meta-analysis of surveys on science fraud (Fanelli 2009)

- 2% admitted to falsifying studies at least once
- 14% admitted to personally knowing someone who did

Replication Crisis

Replication Crisis

The Replication Crisis is an ongoing methodological crisis primarily affecting parts of the social and life sciences beginning in the early 2010s.

- ▶ growing awareness of the problem that results of many scientific studies are difficult or impossible to replicate/reproduce.
- ▶ reproducibility is an essential part of the scientific method,
- ▶ inability to replicate the studies of others has potentially grave consequences for many fields of science in which significant theories are grounded on unreproducible work.

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Observations

(Dis)trust in Science

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By Gleb Tsipursky on July 5, 2018

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More social science studies just failed to replicate. Here's why this is good.

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Uncover Content / The Replication Crisis in Psychology

The Replication Crisis in Psychology

By Edward Diener and Robert Biswas-Diener
University of Utah, University of Virginia, Portland State University

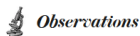
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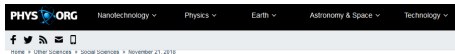
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Home // Monitor on Psychology // 2015 // 10 // A reproducibility crisis?

A reproducibility crisis?

The headlines were hard to miss: Psychology, they proclaimed, is in crisis.

October 2015, Vol 46, No. 9
Print version: page 39



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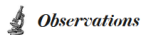
Researcher discusses the the science replication crisis

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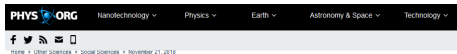


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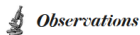
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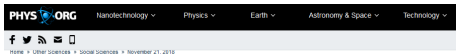
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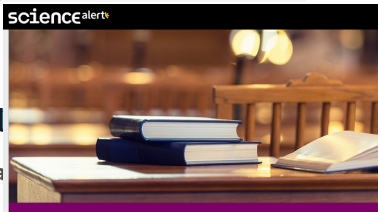
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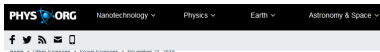
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HUMANS

Science's 'Replication Crisis' Has Reached Even The Most Respectable Journals, Report Shows

MIKE MCRAE 27 AUG 2018



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Our 2019 March Madness

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Psychology's Replication Crisis Has Made The Field Better

By [Christine Aschwendt](#)

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Unravel Controversy / The Replication Crisis in Psychology

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Repercussions

As a consequence of the RC, there is growing awareness...

- of a problem: currently most research is difficult to replicate/reproduce!
- that reproducibility is an essential part of the scientific method
- that the inability to replicate has potentially grave consequences as significant theories are grounded on unreproducible work
- that there is substantial loss of trust in science, its results, and its proponents.

Best Practices in Corpus Linguistics

Best Practices in Corpus Linguistics

As a community, we endorse *blind peer-review*, we are *open to sharing* (if we are asked), and we have *begun with a discussion around best practices* and replication (Berez-Kroeker et al. 2018; Ruhi et al. 2014).

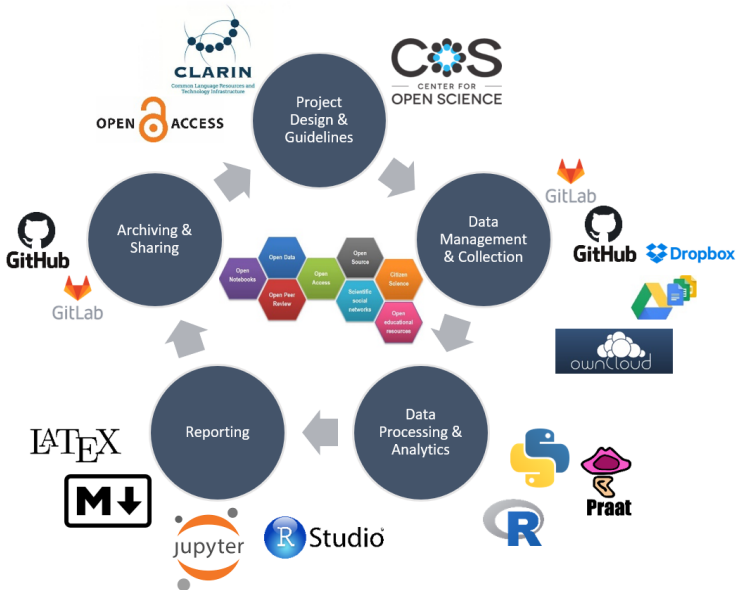
However, we could be better because *our analyses are not reproducible*, we have an *over-reliance on tools*, and *reproduction is discouraged*

- (i) journals are not interested in publishing the same analysis twice;
- (ii) researchers fear repercussions if they criticize the research of others (face-threatening).

Best Practices in Corpus Linguistics

While replicability has improved with the rise of natural language corpora, **we just do not know how bad our research is** (mistakes in using statistical methods or data processing, outright forgery, data manipulation, p-hacking, etc.) because ...

1. researchers do not (or only rarely) reproduce and replicate
2. researchers do not know about best practices or what they are
3. researchers do not know how to make their research comply with best practices
4. lack of training in best practices and how to make research reproducible



Suggestions to make our research more replicable

For individual researchers and teams

- **FAIR principles**

share and make your data ***F**indable, **A**ccessible, **I**nteroperable, **R**eusable* (FAIR) (Wilkinson et al. 2016)

- **Data a publication**

assign a *Digital Object Identifier* (DOI) to your data, provide a clear example for how your data should be cited, and publish it on an online repository (this way your data is a proper publication)

- **Archiving**

use online repositories (e.g. *GitHub*, *GitLab*, *CloudStore*, *MyDrive*, *Dropbox*) to avoid data loss and various versions of a single document or file

For individual researchers and teams

- **Scripts over tools**

use R rather than ready-made software tools because such apps are black-boxes that hinder replication and transparency (due to limited accessibility and/or time-consuming replication)

- **Documentation**

write down what you do and where you store all relevant elements of your project



For individual researchers and teams

- **Folder templates**

think about a schematic folder structure and use it for all your projects, e.g. always using subfolders for *data*, *tables*, and *images* for research projects or *slides*, *exercises*, *student materials*, and *assignments* for courses (ideally implement a policy in your team so that all team members use the same folder template)

- **Notebooks and version control**

make your research fully transparent and reproducible by using R Notebooks and sharing entire projects on GitHub

For the community

Endorse *Open Science*

Open Data + Open Access + Open Methodology + Open Educational Resources

- Ask for data and scripts when reviewing papers
- Cite appropriately (rewards publication of corpora)
- Promote and support replication
- Invest in/support training of staff and students in data management and transparent data analysis options (*R*, *Git*, *Markdown*, *wikis*, etc.)

Continue the discussion and talk to colleagues about *Best Practices/Replication*

Live long and replicate!

Thank you!

Acknowledgements

I would like to thank...

Amanda Miotto
for sharing materials about Replicability in Research
(<https://github.com/amandamiotto/Reproducible-Research-Things>)

my colleagues at UQ
for comments and their feedback on earlier versions of this talk

- Aschwandan, C. (2018). Psychology's replication crisis has made the field better.
<https://fivethirtyeight.com/features/psychologys-replication-crisis-has-made-the-field-better/>.
- Berez-Kroeker, A. L., L. Gawne, S. S. Kung, B. F. Kelly, T. Heston, G. Holton, P. Pulsifer, D. I. Beaver, S. Chelliah, S. Dubinsky, et al. (2018). Reproducible research in linguistics: A position statement on data citation and attribution in our field. *Linguistics* 56(1), 1–18.
- Diener, E. and R. Biswas-Diener (2019). The replication crisis in psychology.
<https://nobaproject.com/modules/the-replication-crisis-in-psychology>.
- Fanelli, D. (2009). How many scientists fabricate and falsify research? a systematic review and meta-analysis of survey data. *PLoS One* 4, e5738.
- McRae, M. (2018). Science's 'replication crisis' has reached even the most respectable journals, report shows.
<https://www.sciencealert.com/replication-results-reproducibility-crisis-science-nature-journals>.
- Resnick, B. (2018). More social science studies just failed to replicate. here's why this is good.what scientists learn from failed replications: how to do better science.
- Ruhi, Ş., M. Haugh, and T. Schmidt (2014). *Best practices for spoken corpora in linguistic research*. Cambridge: Cambridge Scholars Publishing.
- Velasco, E. (2019). Researcher discusses the the science replication crisis.
<https://phys.org/news/2018-11-discusses-science-replication-crisis.html>.
- Weir, K. (2015). A reproducibility crisis? the headlines were hard to miss: Psychology, they proclaimed, is in crisis. *Monitor on Psychology* 46, 39.
- Wilkinson, M. D., M. Dumontier, I. J. Aalbersberg, G. Appleton, M. Axton, A. Baak, N. Blomberg, J.-W. Boiten, L. B. da Silva Santos, P. E. Bourne, et al. (2016). The fair guiding principles for scientific data management and stewardship. *Scientific data* 3. <https://www.nature.com/articles/sdata201618>.
- Yong, E. (2018). Psychology's replication crisis is running out of excuses. another big project has found that only half of studies can be repeated. and this time, the usual explanations fall flat.
<https://www.theatlantic.com/science/archive/2018/11/psychologys-replication-crisis-real/576223/>.

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